

Data Inventory & Remaining Gaps

After processing the ESR, CGWB and basin action plan

⚠️ First: two claims from earlier documents are now dead

The new data overturns them. Both would have been caught on stage.

1. "265 MLD flows untreated into the rivers" — wrong, and badly out of date

That was a 2011 figure. The NEERI Environmental Status Report 2023-24 lists **13 STPs, all operational, totalling 423.5 MLD of installed capacity**, each with coordinates:
Bhandewadi 130 + Bhandewadi 200 + Somalwada-1 20 + Somalwada-2 20 + Narsala 20 + Itabhatti 10 + Mokshdham 5 + Mankapur 5 + Dabha 5 + Hazaripahad 4 + Ambazari 3.2 + Kachimet 1 + Sonagaon 0.3
Nagpur went from 100 MLD of treatment capacity in 2011 to 423.5 MLD in 2024. Saying "265 MLD untreated" in front of a municipal jury would have been both wrong and insulting.

2. "The river doesn't arrive polluted — the city does that to it" — not supported by the current data

The 2023-24 coliform figures don't show that gradient. **Nag-1, the initial sampling point at Ambazari, is the *most* contaminated at 600,000 CFU/100mL total coliform.**

✓ But the real story is better, and it's in the new data

Dissolved oxygen along the Nag, in order downstream:

3.67 → 3.63 → 3.88 → 3.81 → 1.90 → 1.20 → 1.02 → 1.10 → 1.20 → 0.48

A textbook oxygen sag. And the collapse point is named in NEERI's own site description: **Nag-5 is "drain mix behind Yashwant Stadium."** The oxygen dies exactly where the drains join. Same pattern on the Pili (4.20 → 0.74) and the Pora (4.80 → 0.63).
21 of 23 river sites fail CPCB Class D (DO ≥ 4 mg/L — the minimum for fish and wildlife). By its end point the Nag is at 0.48 mg/L: biologically dead water.

The narrative this supports — and it's far stronger than the old one

Nagpur more than quadrupled its sewage treatment capacity between 2011 and 2024. The rivers are still dead. Because the problem was never treatment capacity — it's that sewage enters the rivers before it ever reaches a plant. You can't treat water you never collected.

This gives the city full credit for what it built, and locates the actual failure precisely. That is a far better thing to say to a municipal jury than an accusation — and the DO curve proves it in one chart.

And the single most striking finding in all of this

All 12 city groundwater sources NEERI sampled show coliform contamination — total coliform 46–164 CFU/100mL, thermotolerant faecal 10–80. IS 10500 requires **zero**.

Twelve out of twelve wells sampled inside Nagpur city fail the drinking water standard for faecal contamination.

That's current, primary, government-commissioned, and it's the sharpest single sentence available to you.

What you now have

Dataset	Contents	Status
CGWB district groundwater	347 samples (263 shallow, 84 deep), coordinates, pH/EC/TDS/TH/Ca/Mg/Na/K/HCO ₃ /Cl/SO ₄ /NO ₃ /F/ SAR/RSC	✓ Extracted, verified against report statistics exactly
NEERI rivers 2023-24	23 sites (10 Nag, 8 Pili, 5 Pora), coordinates, full chemistry + coliforms + DO/COD/TKN/phosphate	✓ Extracted
NEERI city groundwater	12 wells , coordinates, chemistry + metals + coliforms	✓ Extracted
NEERI lakes	10 lakes , chemistry + nutrients + coliforms + plankton	✓ Extracted
STPs	13 plants , coordinates, capacity, utilisation, technology, year	✓ Extracted
Non-revenue water	Month-wise supply FY23-24: 257.6 MCM supplied vs 410 MLD billed	✓ Extracted
Outfalls	9 Nag + 3 Pili, named locations	✓ Named, need geocoding
OSM waterways	89 drains, 135 streams, 54 rivers, 24 canals	✓ Exported
ESR command-area maps	Per-reservoir maps with supply zones and supply hours	🗺 Maps, not tables — see gaps

Delivered files: `nagpur_groundwater_cgwb.csv` (347 rows, with IS 10500 and CPCB irrigation flags pre-computed) · `nagpur_neeri_2023-24.geojson` (48 mapped features) · `nagpur_neeri_tables.zip` (15 raw tables)

What you're still missing — collect broadly, decide later

Tier 1 — would materially improve the product

- ESR command-area boundaries as polygons.** You have the maps as PDFs. They show reservoir boundaries, supply zone numbers (SZ-8, SZ-9...) and **supply hours** — which is exactly the "24x7 vs intermittent" variable that the peer-reviewed Nagpur research links to contamination. But they're page images, not data. **Someone traces the boundaries by hand in geojson.io** — roughly 2 hours for the main ones. This unlocks F1 properly instead of the zone fallback.
- Download the rest of the ESR supply maps.** Your scraper targets the right page. There are more than the 8 you've pulled — get all of them, and note the supply-hours legend on each.
- Tap water quality with any spatial resolution.** Still the central gap, and still the thing your problem statement is *about*. OCW publishes only a city-wide fitness percentage. Worth one email on the 16th — and worth saying on stage that you asked.
- NMC ward and zone boundaries as GeoJSON.** Needed for the fallback in F1 and for any per-area aggregation. Check the Nagpur district site and any open-data portal.

Tier 2 — strengthens specific features

- Historical NEERI reports (ESR 2022-23, 2021-22).** Same tables, earlier years → **trend**. One year is a snapshot; three years is a story about whether things are improving. This is the highest-value item in Tier 2 by a distance, and it's the same source you already know how to parse.
- Lake sampling locations with coordinates** — you have the lake chemistry, but the location table needs pulling out properly (Table 6.4).
- STP status from the MPCB NRCD page** you found. Cross-checks the STP list and may add under-construction plants.
- Nagpur district disaster management page** — flood-prone area lists, control room data, historical incident records. This is the honest source for Q2 seed data.
- Rainfall.** Open-Meteo for forecast, IMD for historical daily. Needed for any Q2 rainfall-response work.
- The two academic papers** — groundwater near the Nag river, and the 2026 surface water assessment. Both likely contain sampling tables that add points and, more usefully, independent corroboration of NEERI's findings.

Tier 3 — nice to have, don't chase

- r/nagpur harvest** for flood and water complaint locations.
- Street View historical imagery** for before/after on works.
- Sentinel-1 flood extent** for Sept 2023.
- DEM** — only needed if Q2 goes ahead.

Data you will not get, so stop looking

- Per-outfall discharge volumes** — do not exist publicly. Attribution stays segment-level.
- NMC's internal desilting schedule** — not public.
- Reservoir-level quality results** — this absence *is* your problem statement.

What changes in the build

The bacteriological gap is now partly closed. You have coliform counts for rivers, lakes and 12 city wells. So the verdict engine can assess IS 10500's bacteriological requirement **at those points** — including the finding that all 12 fail it.

But the asymmetric verdict still stands. You have coliforms for 45 specific points, not for anyone's tap. The three states remain — *exceeds limit / no exceedance in the parameters tested here / not tested* — and you still never say "safe."

The irrigation feature is stronger than expected. The CGWB data carries **SAR and RSC** alongside EC and pH, which is the full FAO-29 irrigation assessment rather than the partial one I'd assumed.

A new candidate for your best visual: the DO curve down each river, with the drain-confluence points marked. It's one chart, it's from a government report, it shows cause and effect, and it needs no explanation.